RAMAIAH UNIVERSITY OF APPLIED SCIENCES

Faculty of Engineering and Technology
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Campus Flow AI: Intelligent Document Request and Approval Framework

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Introduction

- Processing official document requests in universities is often a time-consuming and manually intensive task. It involves collecting student details, validating information, forwarding documents through multiple levels of approval, and obtaining authorized signatures.
- Since much of the process is handled manually, it can lead to errors, repeated corrections, misplaced documents, and delays in approvals. The lack of real-time tracking also reduces transparency for students and increases administrative workload.
- To address these challenges, we propose a smart digital request and approval system that automates submission, validation, routing, and document generation. The system integrates Role-Based Access Control for secure interaction and uses an Agentic AI–based decision engine to intelligently evaluate requests and route them efficiently.
- By digitalizing the workflow and incorporating intelligent automation, the system improves efficiency, reduces errors, and ensures secure and transparent document processing.

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Literature Survey

Ref.	Title and Year	Methodologies	Dataset if applicable	Limitations	Observations
[1]	Implementation of e-office applications at Universitas Negeri Malang (2023)	Qualitative case study; interviews, document analysis, observation	Administrative staff and internal documents at Universitas Negeri Malang	Single-institution case; limited generalizability; self-reported data bias	E-office improved efficiency, document traceability, and workflow transparency; change resistance and infrastructure gaps noted
[2]	Development of a workflow management system for higher education (2023)	System design and development; prototype implementation; functional testing	Departmental workflow processes at Fountain University Osogbo	Department-level implementation; limited scalability validation	Automation reduced delays, improved tracking, and enhanced administrative coordination
[3]	Enhancing university operations: Study of EDMS (2024)	Case study; surveys; interviews; system evaluation	Users of EDMS within one higher education institution	Limited longitudinal evaluation; single-institution scope	EDMS enhanced retrieval speed, compliance, and operational efficiency; user training critical
[4]	Drivers of Digital Signature System Adoption (2022)	Quantitative survey; Structural Equation Modeling (SEM)	Finance and Information Systems departments	Cross-sectional design; department-specific context	Perceived usefulness, ease of use, and organizational support significantly influence adoption

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Literature Survey

Ref.	Title and Year	Methodologies	Dataset if applicable	Limitations	Observations
[5]	Implementation of e-Office in R&D Institutions of Power Sector (2023)	Quantitative survey; statistical attitude analysis	Employees in R&D institutions (India power sector)	Sector-specific; focuses on perception over performance metrics	Positive attitudes linked to digital literacy; resistance due to skill gaps and training limitations
[6]	Digital transformation in the management of higher education institutions (2025)	Mixed-methods; strategic analysis; literature synthesis	Multi-institutional cases and secondary data	Broad scope; limited empirical depth per institution	Leadership, governance, and culture drive transformation; funding and legacy systems are key barriers
[7]	Transforming higher education with RPA (2025)	Conceptual framework; case illustrations; application modeling	Case examples from higher education institutions	Emerging technology; limited large-scale empirical validation	RPA reduces repetitive workload, improves efficiency, and supports sustainability goals
[8]	Development of e-document management system for university (ISO 9001:2015 alignment) (2024)	System Development Life Cycle (SDLC); prototype deployment	University administrative processes aligned with ISO 9001:2015	Short evaluation period; compliance-focused scope	Supports audit readiness, document control, and quality assurance processes

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Literature Survey

Ref.	Title and Year	Methodologies	Dataset if applicable	Limitations	Observations
[9]	Digital conversion of document management at universities (2024)	Case analysis; process mapping; system evaluation	University document management workflows	Context-specific findings; limited comparative data	Digital conversion enhances transparency, speed, and accountability
[10]	Automating higher education administrative processes with AI-powered workflows (2026)	AI-based workflow modeling; system design; pilot implementation	Administrative workflow datasets from HEIs	Preprint; limited long-term validation; governance concerns	AI significantly reduces processing time and enhances decision support; ethical concerns highlighted
[11]	Digital transformation of education: Limitations and opportunities (2024)	Literature review; conceptual analysis	Secondary global education data	Theoretical orientation; limited primary empirical evidence	Digital transformation improves access and innovation but faces inequality and infrastructure barriers
[12]	Student Activity Request Approval Management System (2025)	Web-based system development; database integration; testing	Student approval request workflows	Prototype-level testing; limited scalability study	Reduced approval cycle time; improved transparency and tracking

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Literature Survey

Ref.	Title and Year	Methodologies	Dataset if applicable	Limitations	Observations
[13]	Web-based placement/workflow management system (2025)	Software development; database management; system testing	Student placement and administrative records	Project-level implementation; limited cross-institutional validation	Centralized workflow enhanced coordination among students, faculty, and employers
[14]	Case study on implementation of eOffice in Mantralaya (2023)	Government implementation case study; performance analysis	State government administrative departments	Public-sector specific; limited academic institutional comparison	Strong leadership and infrastructure investment critical for large-scale success
[15]	Transforming EDMS adoption and user experience in higher education: Systematic review (2025)	Systematic literature review (2022–2025); thematic synthesis	Peer-reviewed journal and conference papers	Dependent on available literature; potential publication bias	Adoption influenced by usability, organizational readiness, policy support; need for long-term impact studies

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Problem Statement

- Despite ongoing digital transformation in educational institutions, the process of requesting official documents such as bona fide certificates, transcripts, and permissions often remains manual and inefficient. Students are required to make multiple visits, lack real-time tracking, and experience delays due to manual validation and approval procedures.
- Therefore, there is a need to develop a smart, secure, and intelligent request-and-approval system that supports online submission, real-time status tracking, role-based authorization, and AI-driven workflow routing with digital signing and verification mechanisms.

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Objectives

- **Objective 1**:To analyze stakeholder requirements and design a structured data acquisition mechanism .
- **Objective 2**: To design and implement a secure digital platform that enables online submission and processing of official document.
- **Objective 3**: To develop and integrate an Agentic AI–based decision-making engine .
- **Objective 4**: To implement secure document generation with digital signing.

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Sustainable Development Goals(SDG) Aligned to project

SDG No.	Goal Name	Specific Target / Section	Description
16	Peace, Justice and Strong Institutions	16.6 – Develop effective, accountable and transparent institutions at all levels.	The proposed system strengthens institutional governance by ensuring transparency, accountability, and secure administrative workflows. Role-Based Access Control, audit logging, digital signatures, and QR-based verification prevent document forgery and enhance trust in institutional processes.
9	Industry, Innovation and Infrastructure	9.c-Increase access to information and communications technology.	The project applies AI and The project modernizes traditional administrative systems by integrating Agentic AI, intelligent routing mechanisms, and secure digital infrastructure. It promotes innovation and digital transformation within educational institutions.

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Methods

Objective	Method/ Methodology	Description
OB1	Requirement Analysis and Structured Data Acquisition Design	Identifies stakeholder requirements (students, managers, authorities) and designs structured input forms, validation rules, and database schemas to ensure accurate and organized data collection for document requests.
OB2	AI Scheduling Techniques (Backtracking, Forward Checking, Constraint Propagation, Heuristic Search)	Develops a layered web-based system where user roles are defined and permissions are assigned to ensure secure, structured, and authorized interactions during submission, verification, and approval processes.
OB3	Agentic AI-Based Decision Engine with confidence Scoring and Intelligent Routing.	Implements an AI-driven workflow mechanism that evaluates request completeness, computes a confidence score, and dynamically routes requests to managers or authorities based on predefined thresholds.
OB4	Secure Document Generation Digital Signing and System Testing	Generates automated PDF documents from approved requests, applies digital signatures, embeds QR codes for verification, and conducts functional and security testing to ensure authenticity, reliability, and system performance.

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Methods and Methodology

Objective 1: To analyze stakeholder requirements and design a structured data acquisition mechanism for collecting and managing official document request information.

Methods:

- Requirement Analysis
- Structured Data Modelling
- Input Validation Framework
- Database Schema Design

Methodology:

• Identify stakeholders (students, managers, authorities) and document their functional requirements.
• Design structured input forms with mandatory fields and validation rules.
• Model database tables for users, requests, approvals, and audit logs.
• Implement data validation checks to ensure completeness and correctness before processing.

Justification:

• Structured data acquisition ensures accuracy and prevents incomplete submissions.
• Proper database modelling maintains data integrity and traceability.
• Validation reduces manual errors and rework.

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Methods and Methodology

Objective 2: To design and implement a secure digital platform that enables online submission and processing of official document requests while incorporating Role-Based Access Control (RBAC) to ensure structured and authorized interaction among students, managers, and approving authorities.

Methods:

- Layered System Architecture
- Role-Based Access Control (RBAC)
- REST API Development
- Authentication & Authorization (JWT)

Methodology:

- Develop a web-based platform using layered architecture (UI, Application, Data layers).
- Define user roles (Student, Manager, Authority, Admin).
- Assign permissions to each role to restrict unauthorized actions.
- Implement secure login and token-based authentication.

Justification:

- RBAC ensures controlled and secure access to system resources.
- Layered architecture improves maintainability and scalability.
- Secure authentication prevents unauthorized document approvals.

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Methods and Methodology

Objective 3: To develop and integrate an Agentic AI–based decision-making engine that evaluates submitted requests, assigns confidence scores, and dynamically routes them through the appropriate approval workflow.

Methods:

- Agentic AI Workflow Orchestration
- Confidence Score Computation
- Rule-Based Decision Engine
- Intelligent Routing Mechanism

Methodology:

- Extract relevant request details from submitted forms.
- Compute a confidence score based on completeness, validity, and predefined rules.
- If score ≥ threshold, route directly to authority; otherwise, forward to manager.
- Log AI decisions for transparency and auditing.

Justification:

- AI-driven routing reduces manual workload.
- Confidence scoring enables intelligent and faster decision-making.
- Human-in-the-loop verification ensures reliability.

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Methods and Methodology

Objective 4: To implement secure document generation with digital signing, download functionality, and system testing to ensure authenticity, reliability, and overall system performance.

Methods:

- Automated PDF Generation
- Digital Signature Integration
- System Testing & Validation

Methodology:

- Fetch approved request data and populate predefined document templates.
- Apply an authorized digital signature to ensure authenticity.
- Generate a unique certificate ID or embed a QR code for verification.
- Conduct functional, security, and performance testing.

Justification:

- Digital signing ensures document authenticity.
- Testing guarantees reliability and system stability.

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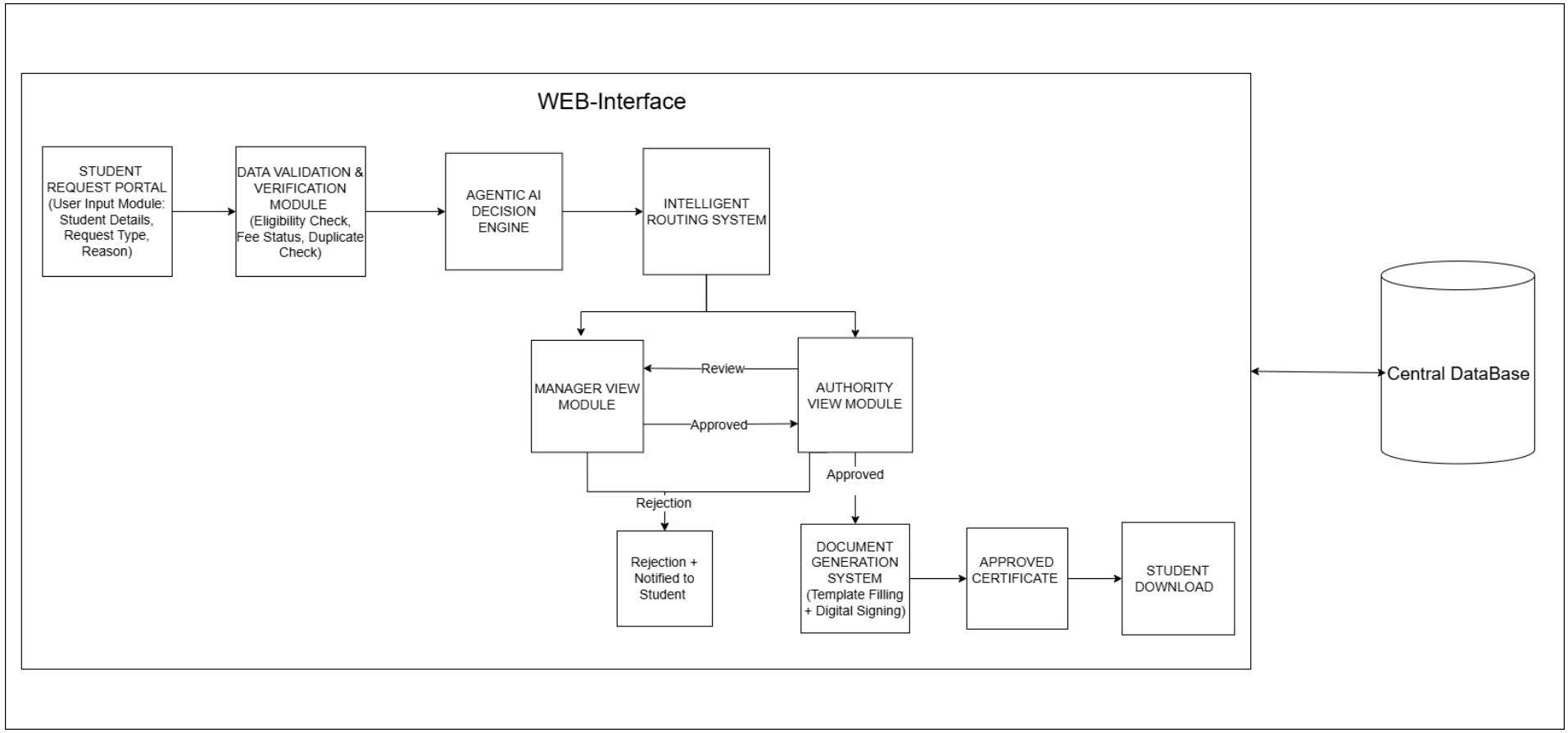
Block Diagram



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Expected Outcomes

- Automated AI-based request and approval system for efficient document processing.
- Secure online submission, tracking, and structured approval workflow using Role-Based Access Control (RBAC).
- Intelligent routing of requests through an Agentic AI–based confidence scoring mechanism.
- Reduced processing time and administrative workload compared to manual procedures.
- Improved transparency, reliability, and accuracy in institutional document management.
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Gantt Chart

Project Work (UG) 16 weeks																
Week	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Major Activities																
1.1 Requirement Analysis & Stakeholder Study																
1.2 Data Acquisition & Database Design																
2.1 System Architecture & Platform Design																
2.2 Backend Development & RBAC Implementation																
3.1 Agentic AI Design & Confidence Score Model																
3.2 AI Workflow Routing & Integration																
3.2 Database Integration & Backend Linking																
4.1 Document Generation & Digital Signing Module																
4.2 System Testing, Evaluation & Documentation																

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Conclusion

- The project addresses inefficiencies, delays, and lack of transparency in manual document request and approval processes.
- It proposes a secure digital platform with Role-Based Access Control for structured and authorized workflow management.
- An Agentic AI–based decision engine enables intelligent request evaluation and optimized routing.
- The system is expected to reduce processing time, minimize errors, and improve institutional efficiency.
- The solution is technically feasible using established web technologies and AI-based workflow mechanisms within the proposed timeline.

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Thank You

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